MADAN MOHAN MALAVIYA UNIVERSITY OF TECHNOLOGY

B.TECH. COMPUTER SCIENCE & ENGINEERING / ALLIED BRANCHES

END SEMESTER EXAMINATION (2023-24) -- SEMESTER 5

COURSE: DESIGN AND ANALYSIS OF ALGORITHMS (DAA) (BCS-502)

Time Allowed: 3 Hours Maximum Marks: 70 Marks Paper Code: BCS-502

Roll Number: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] | Total Printed Pages: 2

NOTE: Attempt all Sections. Assume suitable missing data if any. Programmable calculators are NOT permitted.

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SECTION A: Short Answer Type Questions (Attempt ALL questions) [10 x 2 = 20 Marks]

Q1. (a) State Master Theorem for solving recurrences of form T(n) = a T(n/b) + f(n).

Q1. (b) Define optimal substructure and overlapping subproblems in Dynamic Programming.

Q1. (c) What is greedy choice property in algorithm design?

Q1. (d) What is difference between 0/1 Knapsack problem and Fractional Knapsack problem?

Q1. (e) Define P, NP, NP-Complete, and NP-Hard complexity classes.

Q1. (f) What is polynomial-time reduction?

Q1. (g) What is amortization analysis? Name three methods of amortized analysis.

Q1. (h) What is competitive analysis in online algorithms?

Q1. (i) Explain difference between Backtracking and Branch and Bound.

Q1. (j) What is Las Vegas algorithm vs Monte Carlo algorithm?

SECTION B: Analytical & Computational Problems (Attempt any THREE) [3 x 10 = 30 Marks]

Q2. (a) Solve recurrences using Master Theorem: (a) T(n) = 4T(n/2) + n^2, (b) T(n) = 2T(n/2) + n log n, (c) T(n) = 7T(n/2) + n^2.

Q2. (b) Find optimal parenthesization for Matrix Chain Multiplication with dimensions: <10, 20, 30, 40, 30> using DP m-table and s-table.

Q2. (c) Find Longest Common Subsequence (LCS) of strings X = 'BACDB' and Y = 'BDCB' using Dynamic Programming table.

Q2. (d) Solve 0/1 Knapsack problem with capacity W = 8 and items: {(w=2, v=12), (w=3, v=15), (w=4, v=20), (w=5, v=25)} using DP.

Q2. (e) Explain Fractional Knapsack problem using Greedy strategy and write pseudocode with time complexity.

SECTION C: Long Technical & Design Questions (Attempt any ONE from each) [5 x 10 = 50 Marks]

Q3. (a) Explain Huffman Coding algorithm and build optimal prefix tree for characters with given frequencies. [10 Marks]

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(b) Explain Dijkstra algorithm vs Bellman-Ford algorithm for negative edge weights with cycle detection. [10 Marks]

Q4. (a) Explain Floyd-Warshall all-pairs shortest path algorithm with dynamic programming matrix updates. [10 Marks]

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(b) Explain 0/1 Knapsack using Branch and Bound state-space tree with upper and lower bound calculations. [10 Marks]

Q5. (a) Solve N-Queens problem using Backtracking. Draw state space tree for N = 4. [10 Marks]

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(b) Solve Subset Sum problem using Backtracking for set S = {3, 5, 6, 7} and target sum = 11. [10 Marks]

Q6. (a) State Cook-Levin Theorem and prove that Boolean Satisfiability (SAT / 3-SAT) is NP-Complete. [10 Marks]

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(b) Prove that Vertex Cover problem is NP-Complete by reduction from 3-SAT or Independent Set. [10 Marks]

Q7. (a) Explain Strassen matrix multiplication algorithm and derive its recurrence T(n) = 7T(n/2) + O(n^2). [10 Marks]

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(b) Explain randomized Quicksort with worst-case prevention and expected O(n log n) proof. [10 Marks]

EXAMINATION EVALUATION SCHEME & RUBRIC GUIDELINES:

1. Section A: Direct definition, time complexity analysis, formula derivation, or syntax verification.
2. Section B: Step-by-step mathematical proof, trace tables, algorithmic pseudocode, and state diagrams.
3. Section C: Architectural design, full code implementation, edge case handling, and space/time tradeoffs.

***** END OF QUESTION PAPER -- DEVENGERS ACADEMIC VAULT *****